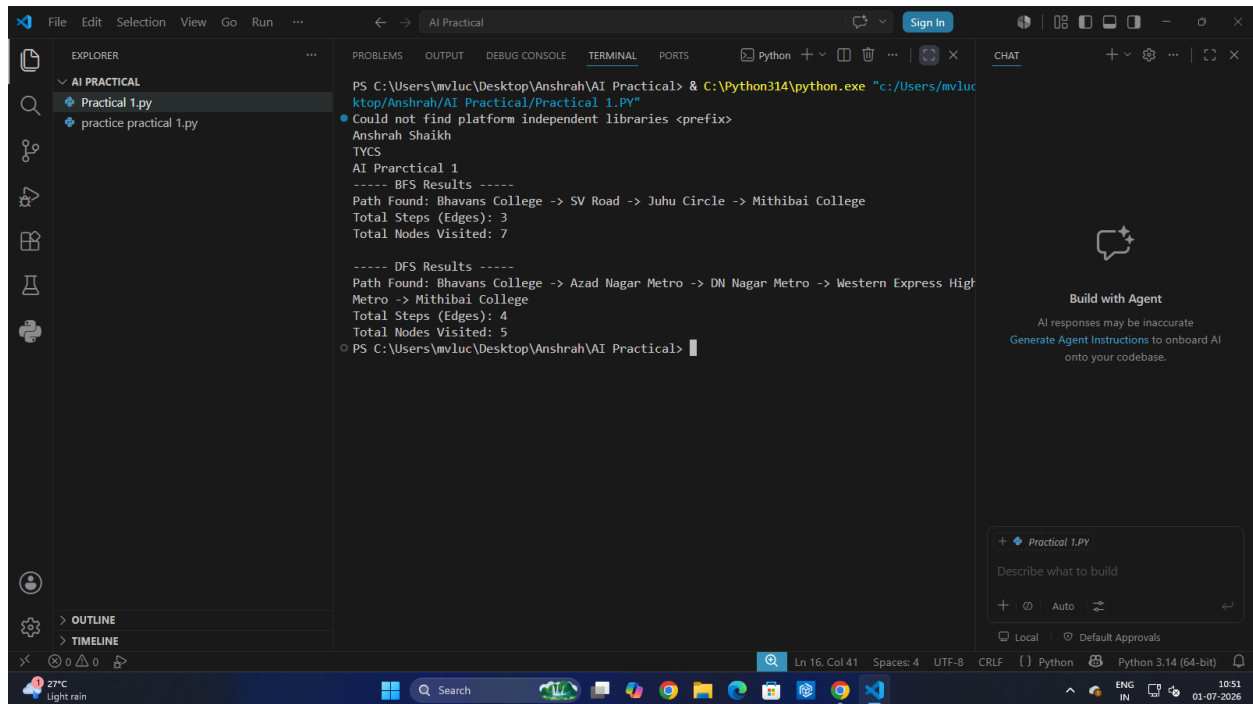


MVLU COLLEGE

Practical 1: Breadth First Search & Iterative Depth First Search in Python.

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

Output:



```
PS C:\Users\mvmluc\Desktop\Anshrah\AI Practical> & C:\Python314\python.exe "c:/Users/mvmluc/Desktop/Anshrah/AI Practical/Practical 1.PY"
Could not find platform independent libraries <prefix>
Anshrah Shaikh
TYCS
AI Practical 1
----- BFS Results -----
Path Found: Bhavans College -> SV Road -> Juhu Circle -> Mithibai College
Total Steps (Edges): 3
Total Nodes Visited: 7

----- DFS Results -----
Path Found: Bhavans College -> Azad Nagar Metro -> DN Nagar Metro -> Western Express High Metro -> Mithibai College
Total Steps (Edges): 4
Total Nodes Visited: 5
PS C:\Users\mvmluc\Desktop\Anshrah\AI Practical>
```

The screenshot shows a Visual Studio Code interface with a terminal window. The terminal displays the output of a Python script. It starts with a command prompt, followed by an error message about missing platform-independent libraries. Then, it prints the name 'Anshrah Shaikh', the course 'TYCS', and the title 'AI Practical 1'. It then shows the results for Breadth First Search (BFS) and Iterative Depth First Search (DFS). The BFS results show a path from Bhavans College to SV Road, Juhu Circle, and Mithibai College, with 3 steps and 7 nodes visited. The DFS results show a path from Bhavans College to Azad Nagar Metro, DN Nagar Metro, Western Express High Metro, and Mithibai College, with 4 steps and 5 nodes visited. The terminal window is titled 'AI Practical' and the file explorer on the left shows 'Practical 1.py' and 'practice practical 1.py'.

ANSHRAH SHAIKH
TYCS
AI PRACTICAL 1